VGG & ResNet

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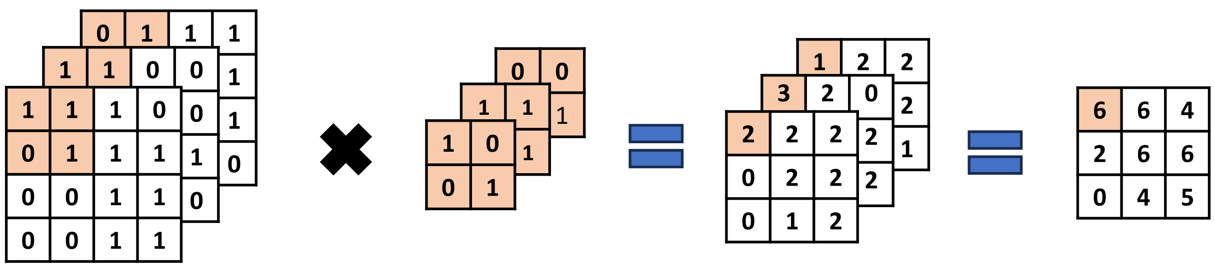
CONTENTS

CNN VGG ResNet 실험 결과

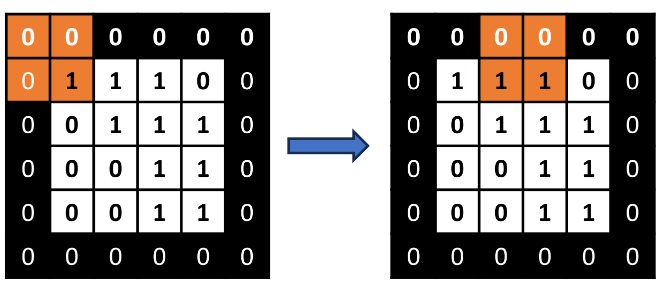
▪ Convolution Layer

- Image classification에 필요한 공간적 특징을 추출한다.

Input Image　Kernel (Filter)　Convolution result　Feature Map

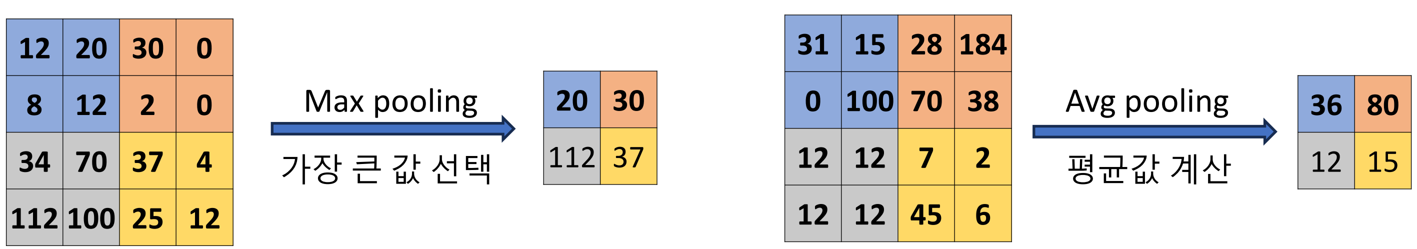


Input image / Padding = 1, stride = 2



$$\frac{W - K + 2P}{S} + 1$$

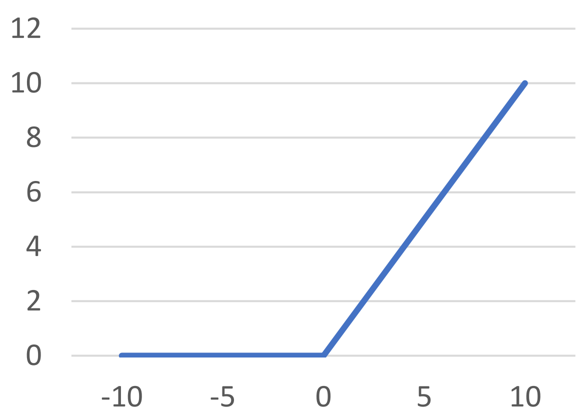
Kernel size와 padding, stride에 대한 Output의 height, width 계산 가능



CNN

▪ Activation Function

Relu



$$R(z) = \max(0, z)$$

3	0	1
-2	0	2
0	2	3

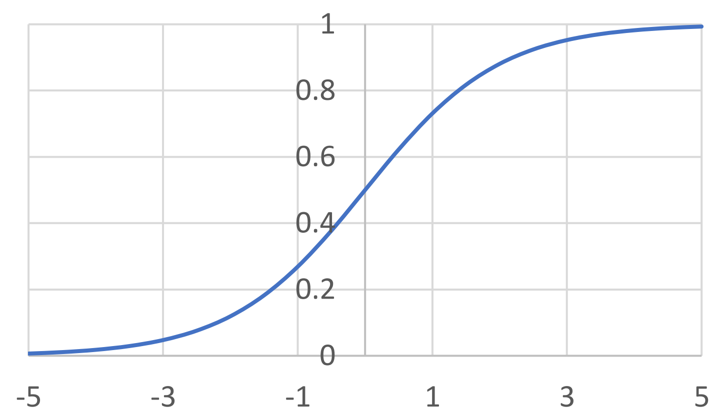
3	0	1
0	0	2
0	2	3

1	-1	1
0	-1	-1
3	1	0

1	0	1
0	0	0
3	1	0

Vanishing gradient 문제점 극복 가능

Sigmoid



$$S(z) = \frac{1}{1+e^{-x}}$$

3	0	1
-2	0	2
0	2	3

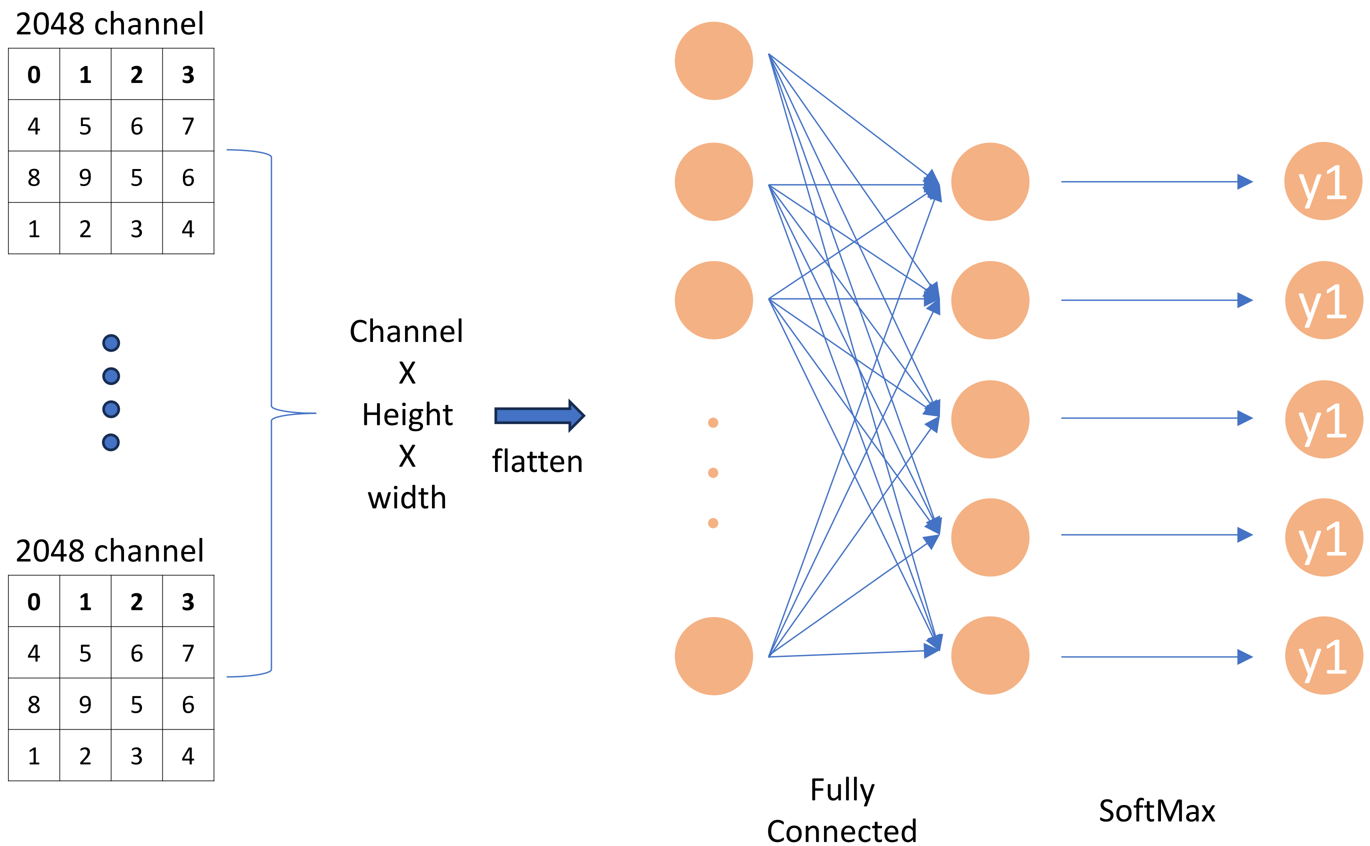
3	0	1
0.12	0	2
0	2	3

1	-1	1
0	-1	-1
3	1	0

1	0.27	1
0	0.27	0.27
3	1	0

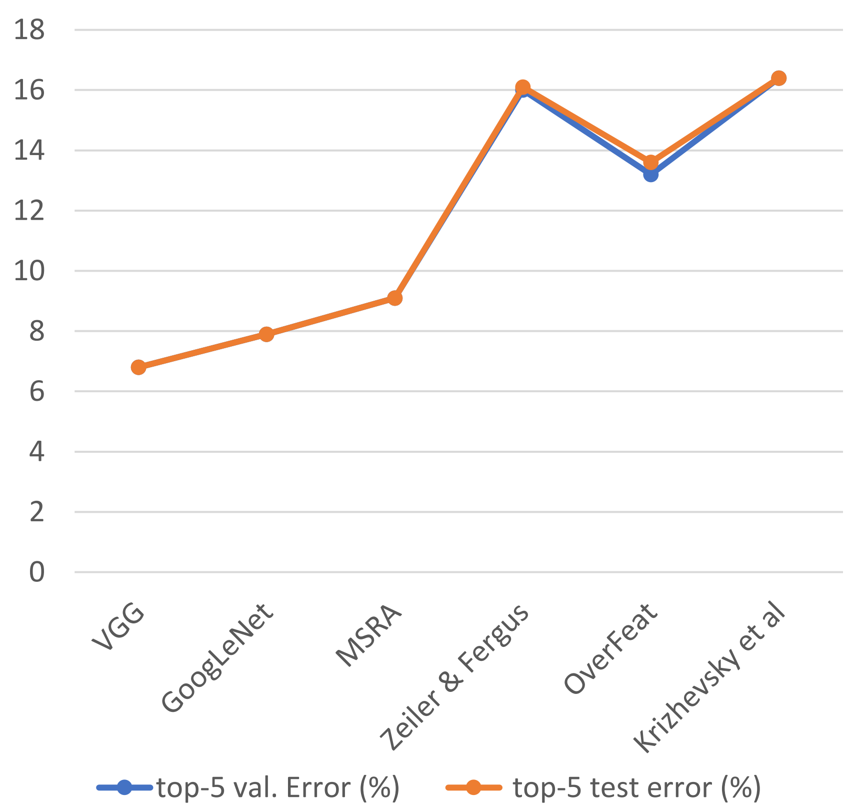
CNN

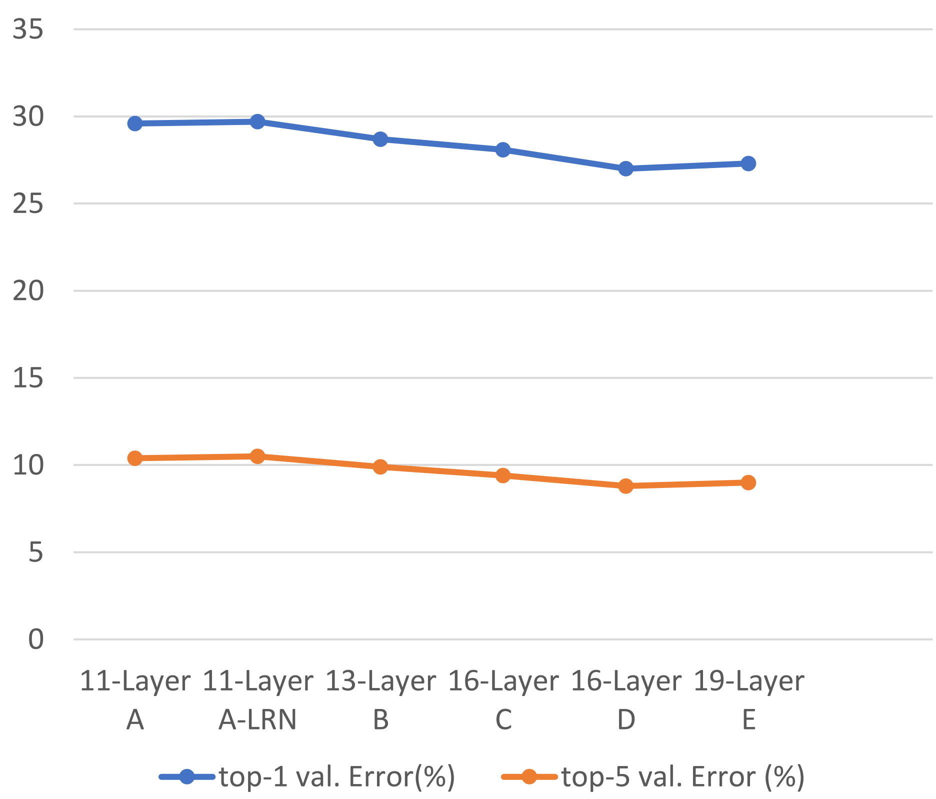
- **Fully Connected**
  - 모든 입력 뉴런이 모든 출력 뉴런과 연결되어 있는 선형 변환 계층



VGG

- **Introduction**
  - CNN의 depth가 large-scale image 인식 정확도에 미치는 영향을 조사
  - ImageNet Challenge 2014에서 Localisation, Classification에서 각각 1,2위 차지

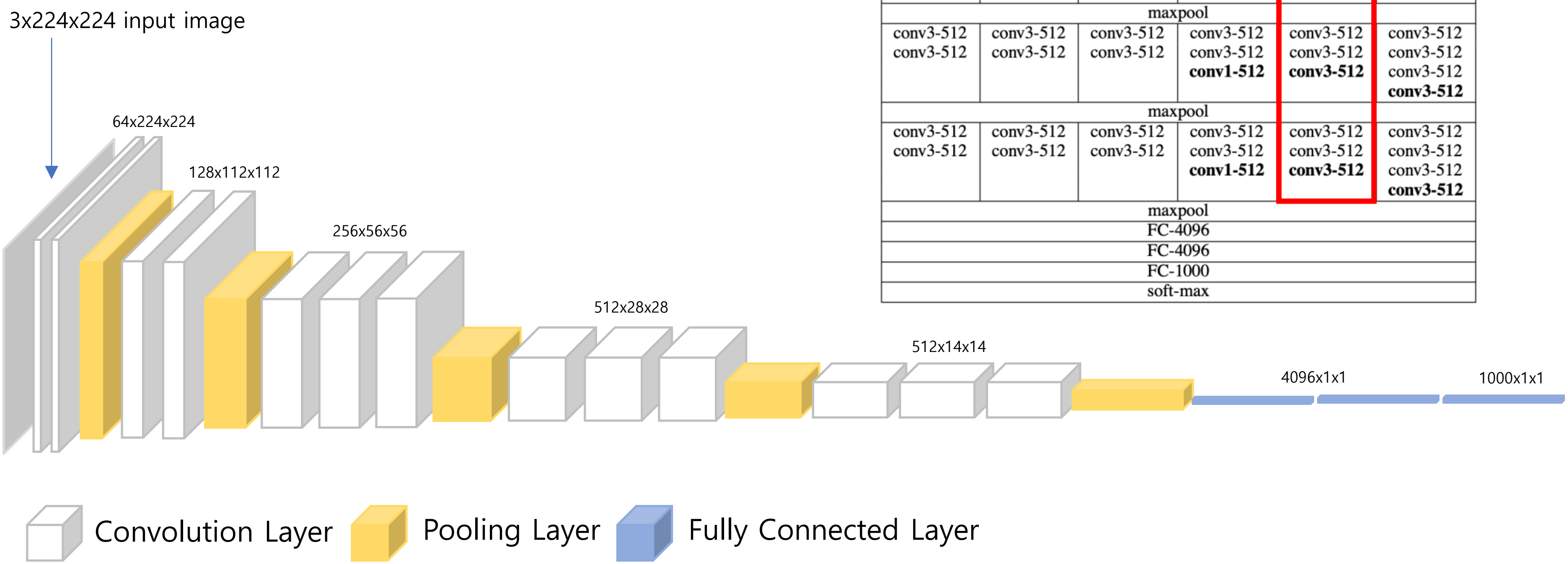




VGG

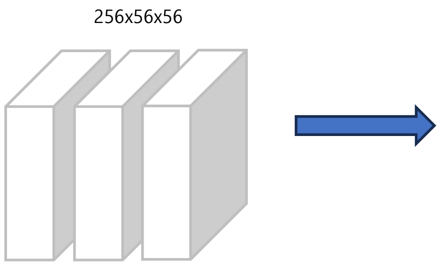
- **architecture**
  - 16 weight layers (D)

ConvNet Configuration					
A	A-LRN	B	C	D	E
11 weight layers	11 weight layers	13 weight layers	16 weight layers	16 weight layers	19 weight layers
input (224 × 224 RGB image)					
conv3-64	conv3-64 **LRN**	conv3-64 **conv3-64**	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64
maxpool					
conv3-128	conv3-128	conv3-128 **conv3-128**	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 **conv1-256**	conv3-256 conv3-256 **conv3-256**	conv3-256 conv3-256 conv3-256 **conv3-256**
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 **conv1-512**	conv3-512 conv3-512 **conv3-512**	conv3-512 conv3-512 conv3-512 **conv3-512**
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 **conv1-512**	conv3-512 conv3-512 **conv3-512**	conv3-512 conv3-512 conv3-512 **conv3-512**
maxpool					
FC-4096					
FC-4096					
FC-1000					
soft-max					

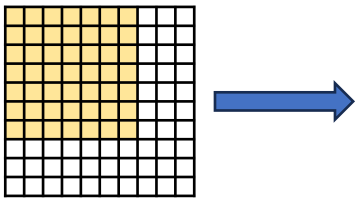


▪ Receptive field, filter

- 3개의 3x3 filter로 하나의 7x7 filter와 같은 receptive field를 가진다.



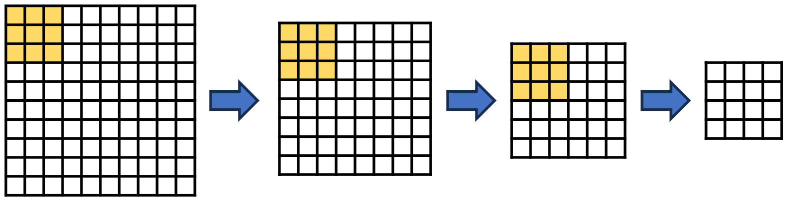
7x7 filter = 7x7 receptive field



Parameter 개수 $(7C)^2 = 49C^2$

3개의 3x3 filter = 7x7 receptive field

- 비선형성 증가, 파라미터 수 감소

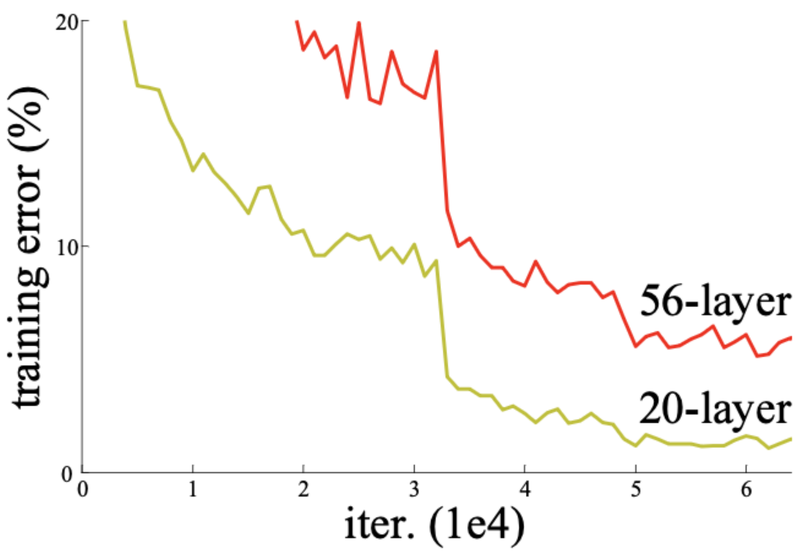


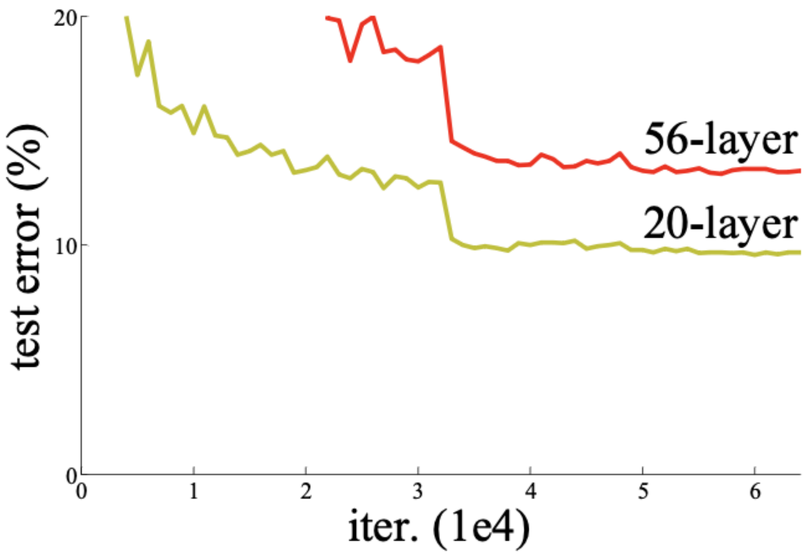
Parameter 개수 $3(3C)^2 = 27C^2$

ResNet

▪ Introduction

- VGG보다 Layer를 훨씬 깊게 했을 때 정확도는 향상하는지 조사
- Ensemble로 test error 3.57% 달성으로 ILSVRC 2015 분류 작업에서 1등 달성





▪ Vanishing / exploding gradiant

- 단순히 더 많은 층을 쌓았을 때 생길 수 있는 장애물이다.

Vanishing gradiant : back propagation 과정에서 출력층에서 입력층으로 갈수록 기울기가 작아지다 0이 되는 현상

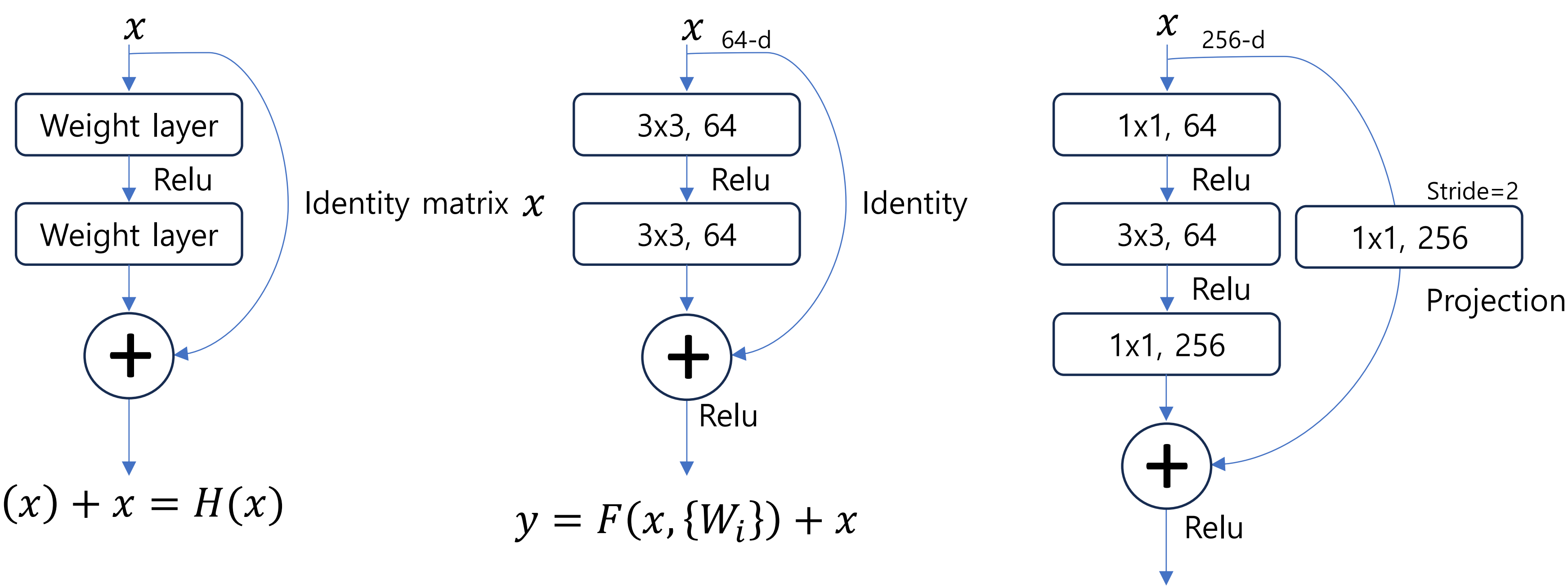
→ Normalized initialization으로 해결 가능

Exploding gradiant : back propagation 과정에서 가중치가 반복적으로 곱해지면서 폭발적으로 증가하는 현상

→ Intermediate normalization Layers으로 해결 가능

▪ Shortcut connection

- 깊은 레이어의 degradation problem을 해결하는데 사용할 수 있다.



$$F(x) + x = H(x)$$

$$y = F(x, \{W_i\}) + x$$

$$y = F(x, \{W_i\}) + W_s x$$

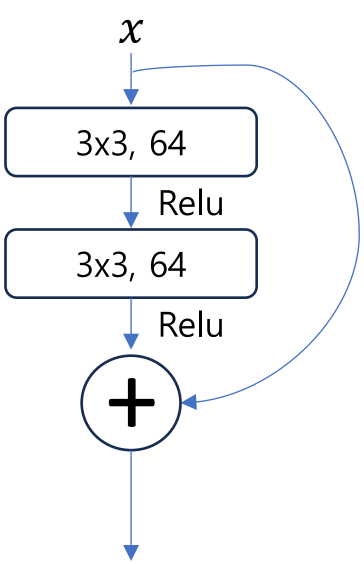
$$F(x) = H(x) - x$$

Residual learning

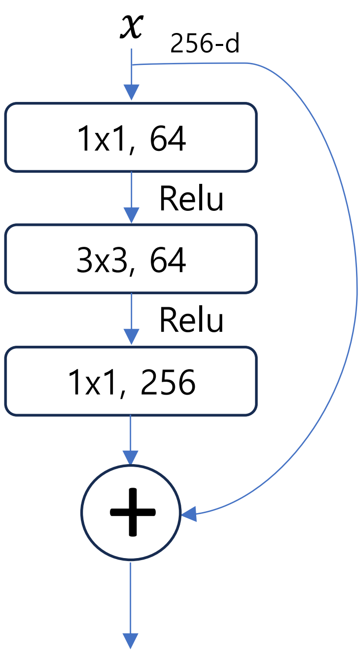
ResNet

▪ Bottleneck architecture

- 연산 효율을 높이기 위해 channel을 일시적으로 줄였다가 다시 확장하는 residual block이다.



$$y = F(x) + x$$

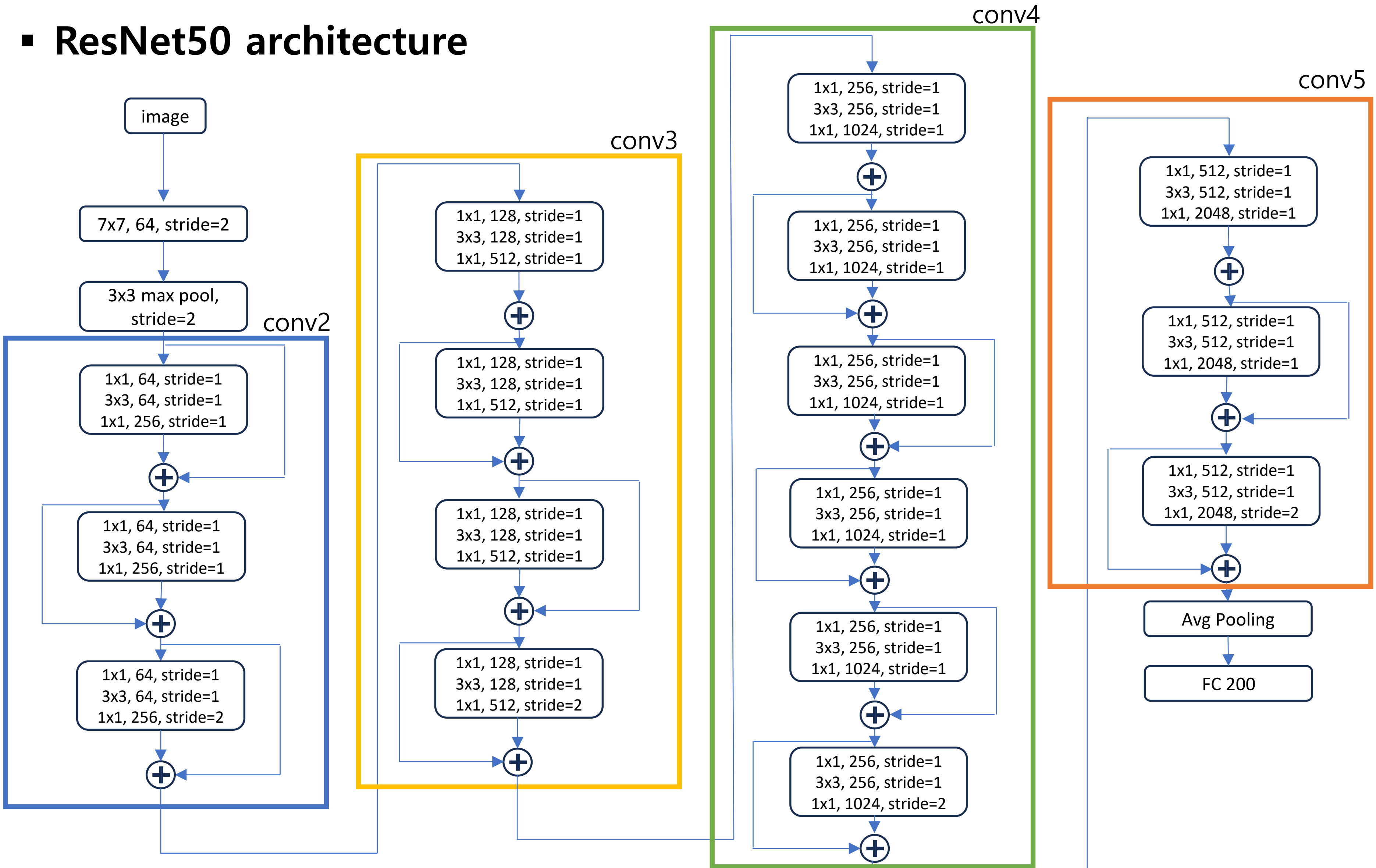


$$y = F(x, \{W_i\}) + x$$

Convolution parameters = kernel width X kernel height X input channel X output channel

ResNet

- **ResNet50 architecture**



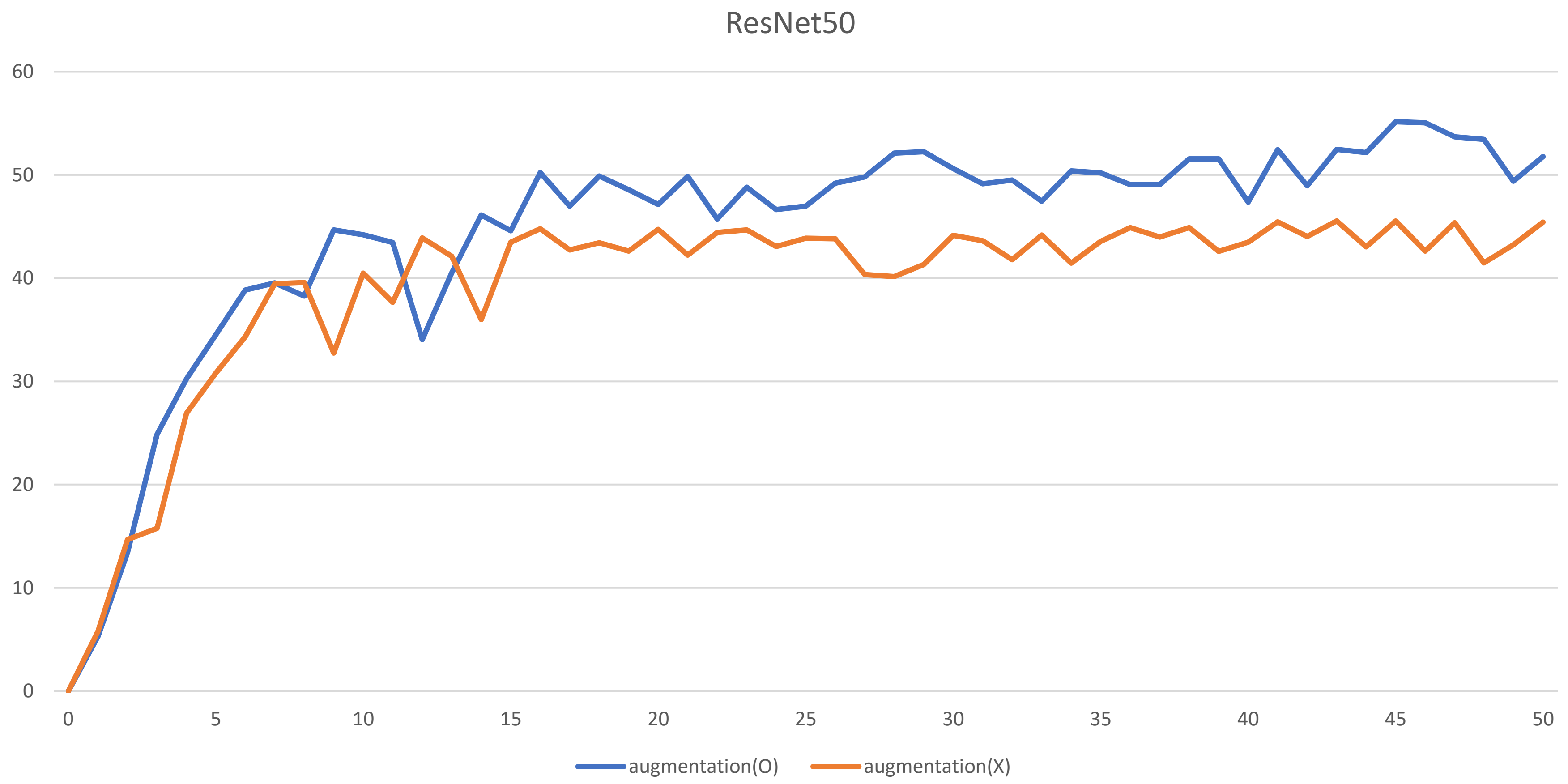
Experiment

▪ ResNet settings

- Dataset
  1. Image : TinyImageSet
  2. Size : 128 x 128
  3. Train : 207,005
  4. Test : 51,752
  5. Class : 200
- Augmentation
  1. Random Crop
  2. Random Horizontal Flip
- HyperParameter
  1. EPOCH : 50
  2. Batch size : 128
  3. Optimizer : SGD
  4. Loss Function : Cross entropy

Result

- **ResNet50**



The end

Thanks

2025. 05. 12